

Fifty Years of Cancer in an American Indian Population

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BACKGROUND: A clear understanding of cancer patterns among American Indian tribal groups has been complicated by a variety of issues. A retrospective cohort study design was applied to a Seneca Nation of Indians (SNI) cohort for the period from 1955 through 2004. **METHODS:** Incident cancers were identified through a computer match with the New York State Cancer Registry. Standardized incidence ratios (SIRs) and 95% confidence intervals were calculated for the overall interval as well as for each of the 5 10-year intervals. The SNI cohort consisted of 3935 men and 4193 women with a total of 120,403 person-years. **RESULTS:** Significant deficits in cancer incidence were noted among men for all sites combined (SIR, 69), and for lung (SIR, 59), prostate (SIR, 54), urinary bladder (SIR, 8), and Hodgkin lymphoma (SIR, 0); no cancer sites were identified with significantly elevated incidence. Women demonstrated significantly reduced cancer incidence for all sites combined (SIR, 70) and for breast (SIR, 39), colorectal (SIR, 72), ovary (SIR, 37), uterus (SIR, 42), bladder (SIR, 20), pancreas (SIR, 10), and non-Hodgkin lymphoma (SIR, 39); elevated incidence was noted for cancers of the lung (SIR, 139) and liver (SIR, 405). **CONCLUSIONS:** To the authors' knowledge, the current study represents the most comprehensive investigation to date of cancer patterns among an American Indian tribal group and provides insights for the development of tribal cancer control programming. **Cancer 2009;115:419–27. © 2009 American Cancer Society.**

KEY WORDS: American Indian, Native American, standardized incidence ratio, incidence.

Despite publications that have described cancer patterns among American Indian populations over the past 20 years,^{1–10} a clear understanding of cancer patterns remains somewhat elusive. Reasons for this include tribal heterogeneity, various sizes of tribal groups, racial misclassification within population databases, aggregation of data for selected tribal groups, and variable time periods examined.

The current report focuses on cancer patterns within the Seneca Nation of Indians (SNI) from 1955 through 2004. This study builds on our earlier work with Seneca elucidating cancer patterns⁴ and cause-specific mortality from 1955 through 1984.^{11,12} To our knowledge, the current study, which covers half a century, represents the most extensive review ever conducted of a single tribe's cancer burden.

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MATERIALS AND METHODS

Study Cohort

The SNI is 1 of the 6 tribes of the Haudenosaunee or Iroquois Confederacy (Mohawks, Oneidas, Onondagas, Cayugas, Senecas, and Tuscaroras) that occupy aboriginal lands in New York State (NYS). The Senecas, the western-most tribe in the confederation, are known as the Keepers of the Western Door.

Although the original territories of the Seneca included vast areas of what is currently NYS, their current territories include the Cattaraugus and Allegany Reservations. The Cattaraugus Reservation is comprised of approximately 22,000 acres overlapping the 3 counties of Erie, Chautauqua, and Cattaraugus in the western-most part of NYS, and the Allegany Reservation extends from the Pennsylvania border northward to Vandalia, New York, and lies entirely within Cattaraugus County. The SNI has a population of >7000 enrolled members, most of whom reside in western NYS. Membership in this matrilineal group is tracked by the Tribal Clerk's office.

Study Design

A retrospective cohort study design was used to examine patterns of cancer incidence among this American Indian population from January 1, 1955 to December 31, 2004 identified as residents of NYS. This is a dynamic cohort with members added through births and removed at the time of death. Membership was based on tribal roll book listings.

Each cohort member contributed 1 person-year, to the appropriate sex and age category for each complete year of survival and 0.5 person-years during their year of birth or year of death. Endpoints for person-year calculations were 1) the date of death (or age 99 years, if the date of death was unknown) or 2) survival thru December 31, 2004. Sex-specific person-year totals for the study cohort were accumulated over 10 5-year intervals (1955-1959, 1960-1964, 1965-1969, ..., and 1995-2004) and were partitioned into 5-year age categories (ages 0-4 years, 5-9 years, ..., 80-84 years, and 85-99 years). Cohort members who developed a malignancy during the study period continued to contribute person-years at risk, allowing for the potential development of multiple primary malignancies.

Person-years at risk among the SNI study cohort were multiplied by corresponding sex-specific, age-spe-

cific, site-specific, and calendar year-specific cancer incidence rates for NYS, exclusive of New York City, for the midpoints of each 5-year interval to yield expected numbers of malignancies during the study period. Cancer incidence rates were based on data from the NYS Cancer Registry (NYSCR) and census enumerations of the NYS population, exclusive of New York City, during corresponding years. Expected cancer cases were summarized across 5 10-year intervals (1955-1964, 1965-1974, 1975-1984, 1985-1995, and 1995-2004).

Incident cancers among the SNI were identified through a computer match of cohort members with data files maintained by the NYSCR during each year of the study period from 1955 to 2004. The matching program that we used was developed and validated by the NYSCR, which actively has collected information on incident cancer cases since 1940, when cancer became a reportable disease in NYS. Matching criteria included last name, first name (first 4 characters), sex, year of birth (± 5 years), and county of residence. Possible matches were checked manually for confirmation with reference to more specific identifying data, such as street address, month, and day of birth; race; or other available information. This manual review allowed for accurate determination of true matches. During the study interval, cancer cases were classified at the NYSCR using various versions of the International Classifications of Diseases (ICD) (eg, the seventh revision [ICD-7], ICD-8, ICD-9, ICD-10, and the ICD-Oncology, third revision). The NYSCR provided information on incident cancers diagnosed among cohort members recorded to the 10th revision of the ICD.¹³

Standardized incidence ratios (SIRs) (based on the ratio of [observed cases/expected cases] multiplied by 100), were calculated for the period from 1955 to 2004 overall and for each of the 5 10-year intervals from 1955 through 2004. Also, 95% confidence intervals (95% CIs) for each SIR were calculated assuming a Poisson distribution.¹⁴ A point estimate of cancer incidence was considered to differ significantly from expected if its 95% CI did not include 100.

RESULTS

The overall SNI cohort consisted of 3935 men and 4193 women who were registered tribal members between 1955 and 2004. Figure 1 presents a population pyramid

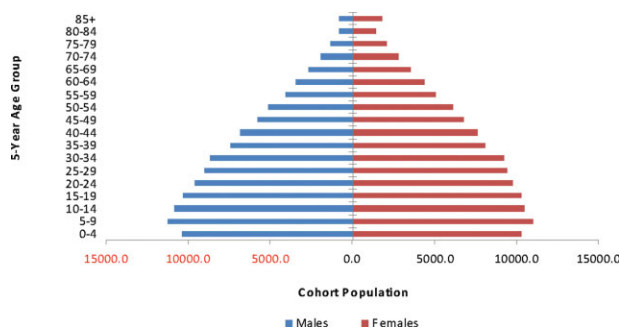


FIGURE 1. Age pyramid for the Seneca Nation of Indians cohort based on person-years accrued from 1955 through 2004 by sex.

based on the 120,403 total person-years of observation accrued among cohort members during the study interval.

Distribution of Incident Cancers

In total, 233 primary malignancies were diagnosed among SNI men between 1955 and 2004. The most frequent malignancy was colorectal cancer ($n = 47$ men; 20%) followed by cancers of the prostate ($n = 39$ men; 17%), lung ($n = 35$ men; 15%), oral cavity ($n = 13$ men; 6%), pancreas ($n = 12$ men; 5%), and kidney ($n = 11$ men; 5%); non-Hodgkin lymphoma ($n = 10$ men; 4%); and leukemia ($n = 10$ men; 4%).

Among women in the cohort, in total, 256 malignancies were observed. Common sites included cancers of the lung ($n = 52$ women; 20%), breast ($n = 41$ women; 16%), colon-rectum ($n = 36$ women; 14%), cervix ($n = 19$ women; 7%), stomach ($n = 10$ women; 4%), and uterus ($n = 10$ women; 4%) and leukemia ($n = 9$ women; 5%).

Standardized Incidence Ratios: 1955 to 2004

Table 1 lists the overall SIR among men for the period 1955 through 2004, which was reduced significantly (SIR, 69; 95% CI, 61-79). Cancer sites/types that exhibited significant deficits included lung (SIR, 59; 95% CI, 41-82), prostate (SIR, 54; 95% CI, 39-74), urinary bladder (SIR, 8; 95% CI, 1-30), and Hodgkin lymphoma (SIR, 0; 95% CI, 0-92). No individual cancer site was identified as significantly elevated.

Women also demonstrated significantly reduced SIRs for all cancer sites combined (SIR, 70; 95% CI, 61-79) (Table 2). Individual cancer sites/types that demonstrated significant deficits included breast (SIR, 39; 95% CI, 28-53), colorectal (SIR, 72; 95% CI, 50-99), ovary (SIR, 37; 95% CI, 14-81), uterus (SIR, 42; 95% CI, 20-77), bladder (SIR, 20; 95% CI, 2-73), pancreas (SIR, 10; 95% CI, 0-58), and non-Hodgkin lymphoma (SIR, 39; 95% CI, 13-91). Elevated SIRs were observed among women for cancers of the lung (SIR, 139; 95% CI, 104-182) and liver (SIR, 405; 95% CI, 175-798).

Standardized Incidence Ratios by Age Group

Cancer incidence patterns by age group are summarized in Table 3. Across the 50-year study period, men demonstrated a deficit of cancer cases for the groups aged 60 to 69 years (SIR, 70; 95% CI, 54-89), 70 to 79 years (SIR, 67; 95% CI, 51-87), and ≥ 80 years (SIR, 59; 95% CI, 40-84). There were no significant excesses observed for any of the age categories.

Women exhibited significant deficits for the groups aged 20 to 29 years (SIR, 26; 95% CI, 3-94), 30 to 39 years (SIR, 55; 95% CI, 28-99), 60 to 69 years (SIR, 78; 95% CI, 60-99), 70 to 79 years (SIR, 54; 95% CI, 39-73), and ≥ 80 years (SIR, 64; 95% CI, 45-88). No significantly elevated SIR was observed for any of the age categories.

Temporal Trends by 10-year Intervals

Figure 2 shows that men exhibited significantly reduced SIRs for overall cancer during 3 of the 5 10-year periods. Although the confidence intervals overlapped, the overall pattern approximated an inverted V. Among individual cancer sites, several notable patterns were evident. For example, whereas lung cancers were absent among men in the first period (SIR, 0), they increased during the third period (SIR, 96), and then declined in the last period (SIR, 52). In contrast, the SIRs across the 5 10-year periods remained relatively stable for prostate cancers and never exceeded 74 for any of the 5 10-year intervals.

Women in the cohort also demonstrated significantly reduced SIRs overall for 3 of the 5 10-year periods. SIRs for breast cancer were significantly low during each

Table 1. Standardized Incidence Ratios for Malignancies Among Men in the Seneca Nation of Indians: 1955-2004

Cancer Site	No. of Malignancies		SIR	95% CI
	Observed	Expected		
All sites	233	335.8	69	61-79
Oral	13	11.3	115	61-197
Esophagus	2	5.8	35	4-125
Stomach	5	9.9	51	16-118
Colorectal	47	46.2	102	75-135
Liver	7	3.4	207	83-426
Gallbladder	2	1	197	24-711
Pancreas	12	9.2	132	68-231
Larynx	7	7.2	97	39-200
Lung	35	59.5	59	41-82
Prostate	39	72	54	39-74
Testis	1	4.7	21	1-118
Bladder	2	24.3	8	1-30
Kidney	11	9.5	115	58-206
Brain	4	6.1	65	18-167
Thyroid	3	2.2	134	28-393
Hodgkin lymphoma	0	4	0	0-92
Non-Hodgkin lymphoma	10	14.1	71	34-131
Multiple myeloma	5	3.9	129	42-300
Leukemia	10	12.4	81	39-149
Other	18			
Total	233			

SIR indicates standardized incidence ratio; 95% CI, 95% confidence interval.

of the last 4 10-year periods, with SIRs ranging between 16 and 55. Other findings of note include an extremely elevated SIR for liver cancers in the last 10-year period (SIR, 663; 95% CI, 215-1548). Trends for the other more common cancer sites exhibited varied patterns.

DISCUSSION

Cancer patterns among American Indian populations have been a topic of debate since the beginning of the 20th century, when reports based largely on clinical observations suggested that American Indian populations were immune from cancer.¹⁵ Observers speculated that this was because of their reliance on an aboriginal diet and natural living, although these differences were more likely because of the younger overall age of American Indian populations. Much has changed in these populations over the past 100 years. In fact, there have been remarkable changes over the past 50 years: The median age of the Seneca cohort increased by 10 years from 24 years in 1955 to 34 years in 2005. Tribal age pyramids over this course of time demonstrate a changing age distribution from being a community with a greater number of

younger individuals to a community with a greater proportion of older individuals. The average age at death among the Seneca in 1955 was 59 years, whereas it was 65 years in 2005.

One of the key challenges in the conduct of epidemiological investigations with Native American populations is that of size. Our work with Seneca remains unique, first because we have been able to maintain trust within the tribe for several decades and to examine disease patterns for an extended period. Concerning trust, prestudy presentations were made to the Tribal Council to gain their mandatory support for this study. Poststudy presentations also were made to the Tribal Council seeking permission to publish and to discuss how study findings would be communicated to the tribal community. Concerning population size, according to the 2000 US Census, >2.5 million American Indians and Alaska Natives identify themselves as members of approximately 570 federally recognized tribes. Only a few of these tribes, such as the Cherokee and the Navajo, are extremely large (>250,000 individuals). The vast majority of tribes have <1000 members, and many have <100. The SNI represents 1 of the largest tribes in the northeastern US. A unique feature

Table 2. Standardized Incidence Ratios for Malignancies Among Women in the Seneca Nation of Indians: 1955-2004

Cancer Site	No. of Malignancies		SIR	95% CI
	Observed	Expected		
All sites	256	368.3	70	61-79
Oral	2	5.9	34	4-121
Esophagus	0	2.1	0	0-175
Stomach	10	6.8	147	70-270
Colorectal	36	50.2	72	50-99
Liver	8	2	405	175-798
Gallbladder	3	2.2	139	29-406
Pancreas	1	9.6	10	0-58
Larynx	2	1.9	107	13-386
Lung	52	37.5	139	101-182
Breast	41	104.4	39	28-53
Cervix	19	11.7	162	98-254
Uterus	10	24	42	20-77
Ovary	6	16	37	14-81
Bladder	2	9.8	20	2-73
Kidney	8	6.4	125	52-246
Brain	2	5.4	37	4-133
Thyroid	5	7	71	23-167
Hodgkin lymphoma	1	3.5	29	1-161
Non-Hodgkin lymphoma	5	12.9	39	13-91
Multiple myeloma	3	3.7	80	17-234
Leukemia	9	10.4	87	40-164
Other	31			
Total	256			

SIR indicates standardized incidence ratio; 95% CI, 95% confidence interval.

Table 3. Standardized Incidence Ratios for Overall Cancer Incidence Among Seneca Nation of Indians by Sex and Age Group

Age Group, y	Men		Women	
	SIR	95% CI	SIR	95% CI
Birth to 9	129	35-329	72	9-259
10-19	63	8-229	109	23-319
20-29	44	9-128	26	3-94
30-39	47	15-109	55	28-99
40-49	88	54-136	87	62-118
50-59	78	57-105	77	58-101
60-69	70	54-89	78	60-99
70-79	67	51-87	54	39-73
≥80	59	40-84	64	45-88
Overall	69	61-79	70	61-79

SIR indicates standardized incidence ratio; 95% CI, 95% confidence interval.

of our study is that northeastern tribal groups are not only small but are underappreciated in the scientific literature. The recently released Report to the Nation on the Status of Cancer¹⁶ (Report to the Nation) provided an update on cancer among American Indians and Alaska Natives.

Although that report is a significant addition to our understanding of cancer occurrence in American Indian populations, it does have some limitations, and data in the report essentially are cross-sectional (1999-2004) and are not necessarily reflective of individual tribal communities within the reported area. For example, the 3 most common cancer sites among men in the SNI, and also among American Indians in the eastern US, were prostate, lung, and colorectum. However, there were differences compared with the general population: In particular, in the Report to the Nation, colorectal cancers reportedly were half as common as in the general population, whereas we observed that they were equally as common (SIR, 102) in our study. Among American Indian women in the eastern US, the Report to the Nation cited breast, lung and colorectal as the most common. These sites also were the most common among Seneca women; but, again there were marked differences compared with the general population. This was most notable for lung and bronchus (risk ratio, 0.74 in the Report to the Nation¹⁶; SIR, 139 among Seneca) and breast (risk ratio, 0.52 in the Report to the

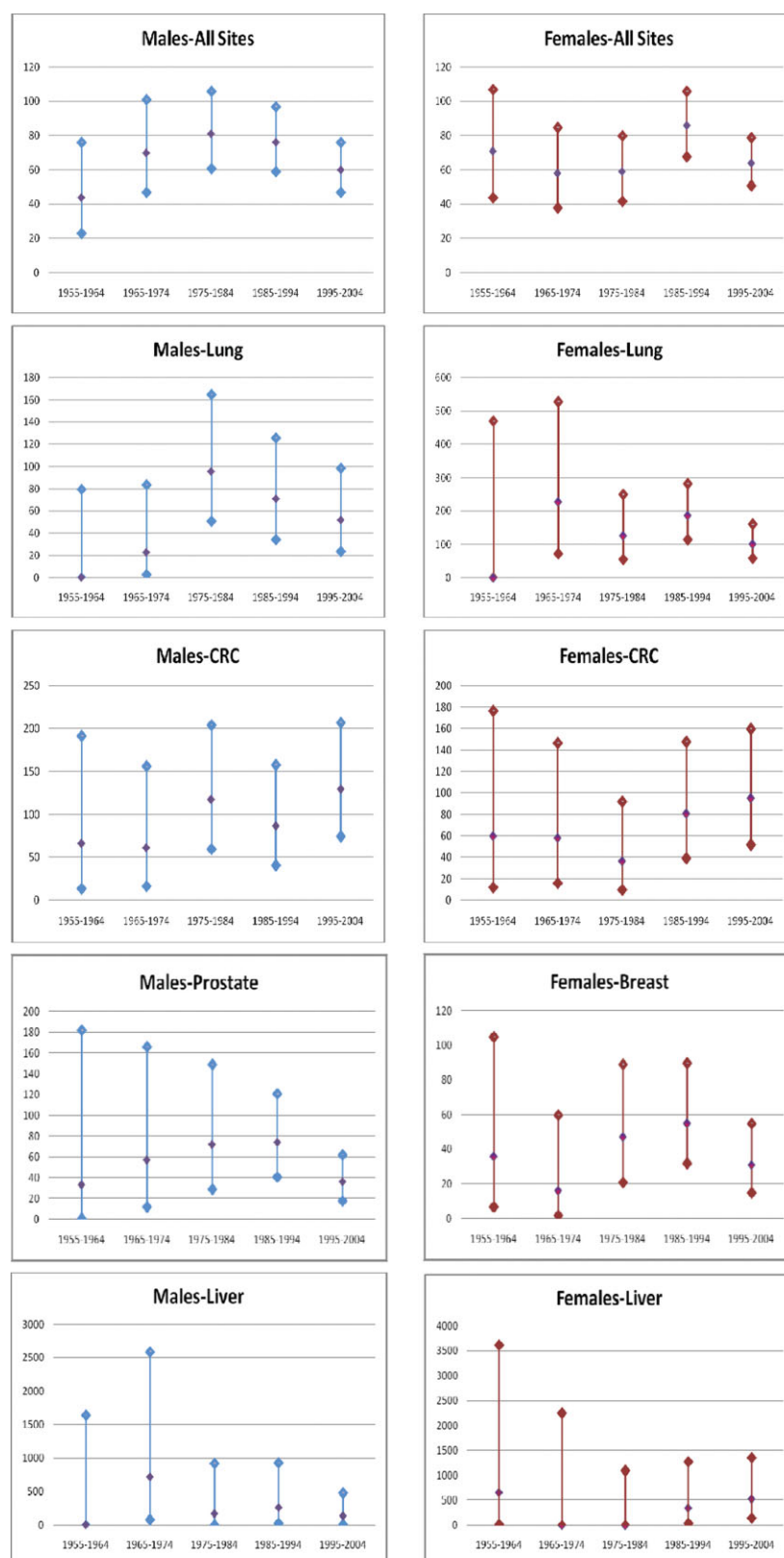


FIGURE 2. Standardized incidence ratios (SIRs) for the Seneca Nation of Indians from 1955 through 2004, by 10-year intervals, overall and for selected cancer sites, by sex. CRC indicates colorectal cancer. The vertical axis corresponds to SIRs; note the different scales for SIRs.

Nation¹⁶; SIR, 39 among Seneca). These nuances underscore the need to work with individual tribal groups to identify their unique cancer patterns, which are of extraordinary value to them in terms of setting health policy, planning activities, and providing them with a unique perspective on disease trends.

Our previous review of cancer incidence patterns among the SNI, based on a fixed cohort of individuals who were alive in 1955, noted significantly lower cancer rates for all sites combined for both sexes; for cancers of the lung and bladder among men; and for cancers of the colon, breast, and bladder among women between 1955 and 1984.⁴ Neither sex demonstrated elevated incidence. The current investigation, which included an additional 20 years of observation among a dynamic population cohort (eg, everyone who was alive in 1955 plus subsequent births), also noted significantly lower incidence ratios for all sites combined among men and women; for lung, prostate, and bladder cancers and Hodgkin lymphoma among men; and for breast, colorectal, uterine, ovarian, and bladder cancers and non-Hodgkin lymphoma among women. Obviously, summary findings across a span of 50 years may mask significant temporal trends. Thus, we explored cancer patterns across the decades.

A review of trends across 5 10-year intervals suggested that these cancer patterns were somewhat dynamic but generally remained unchanged over the 50-year study interval despite some rather significant changes, such as the establishment of 2 health clinics on reservation lands and integration within surrounding communities. In the current investigation, for lung cancer, we observed an overall SIR of 59 (95% CI, 41-82) among men and 139 (95% CI, 104-182) among women during the 50-year study period. During the original investigation (1955-1984), we noted that, especially among men, SIRs increased for lung cancer from zero between 1955 and 1964, to 23 between 1965 and 1974, then to 88 for the period from 1975 to 1984. Although it was our expectation that lung cancer would continue to increase among men, that continued increase was not observed. Rather, lung cancer incidence among men declined during the most recent 20 years (eg, 1985-1994 and 1995-2004). Among women, we observed an elevated SIR for lung cancer between 1985 and 1994 and an SIR for lung cancer that was comparable to what we expected during 1995 to

2004. Therefore, the question naturally arises regarding which factor or factors may be responsible for the declines in men and the increases in women. Although data on smoking rates were unavailable, we noted that the observed patterns are not the result of wholesale changes in smoking prevalence. Rather, we hypothesize that we are observing an effect of competing causes of mortality. In other words, individuals who are at increased risk of developing lung cancer are dying from other malignancies or from other noncancer causes of death before they can develop lung cancer. This hypothesis is consistent with the deficit of cancer cases observed among the SNI for the group aged ≥ 60 years and with Indian Health Service observations¹⁷ that American Indians have a shorter life expectancy than the general population and are 3 times more likely to die before age 45 years. This hypothesis is being evaluated formally in a separate mortality study of the SNI that currently is underway.

Another interesting trend among men was noted for prostate cancer, for which the SIR increased from 33 for 1955 through 1964 to 74 for 1985 through 1994, followed by a precipitous drop to 39 for 1995 through 2004. Again, the prevalence of prostate-specific antigen (PSA) screening in this population is unknown; however, PSA screening in the general population has resulted in marked increases in the detection of early-stage prostate cancers. A review of staging data for the last 20 years (information from earlier years was not available) revealed that 50% of all prostate cancers that were diagnosed during 1985 through 1994 were localized compared with 67% during 1995 through 2004. This may indicate significant under screening by PSA among tribal members.

The observation of lowered SIRs among the SNI over a 50-year interval lacks a definitive explanation. Again, to extend our early argument, it is possible that competing causes of premature deaths from causes, such as injuries/accidents, diabetes, infectious diseases, and cardiovascular disease, among the Seneca may have eliminated the individuals who were likely to go on to develop cancer later in their lives. That explanation would be consistent with the age-specific deficits observed among both SNI men and women for the age categories ≥ 60 years and among women ages 20 to 29 years and ages 30 to 39 years. Currently, we are completing a mortality study among the SNI that may shed light on this hypothesis. In addition, it is noteworthy that, during the latter part of the 20th

century, 30% of all Native American deaths occurred in individuals aged <45 years compared with 10% of deaths for the US All Races population.¹⁷

Although the Seneca represent a rather robust sized tribal population, cancer remains a relatively rare diagnosis, which makes point estimates somewhat unstable, particularly when it is subdivided by sex, cancer site, and time interval. Although the interpretation of these data points must proceed cautiously, investigators are urged to identify opportunities to pool data from studies that rely on a similar design. Unfortunately, the retrospective cohort design that we used in this study has not been used commonly. Alternatively, the cancer experience of tribal groups located in relatively close geographic proximity might be aggregated as a means of overcoming the limited numbers of cases observed within individual tribal groups of small to moderate size.

Differences in cancer incidence patterns observed among the Seneca relative to other American Indian populations is multifactorial and likely reflects differences in lifestyle, environment, genetics, and use of medical care resources. Moreover, the American Indian population throughout the US is quite heterogeneous, and the extent and impact of genetic admixture with non-Indian populations is unclear. It has been hypothesized that American Indian groups across North America share common genetic origins from ancestors who migrated across the Bering Straits from Asia into North American.¹⁸ Consequently, some similarity of cancer patterns might be expected, although behavioral patterns and other factors may explain some of the observed differences.

Given the breadth of health problems faced by all American Indian tribal communities, the extent of resources focused exclusively on cancer prevention and control services is uncertain and most likely limited. Although cancer screening is addressed within the context of routine medical care, community-based approaches to cancer control and surveillance among individuals at high risk for malignancy are difficult to quantify. In addition, although print and other educational materials on cancer topics are available at tribal health centers, their use has not been measured.

These data on cancer incidence patterns among the SNI underscore the importance of renewed efforts at primary cancer prevention through programs aimed at smoking prevention and cessation in combination with

policy changes to promote the adoption of healthy lifestyles through nutrition, exercise, and maintenance of optimal body weight. In addition, secondary prevention, assuring optimal adherence to cancer screening recommendations, will contribute to improvements in survival as well as quality of life.

Limitations of this study include the finding that detailed information on cancer risk behaviors was not available. Likewise, information on lifestyle exposures, environmental exposures, genetic factors, and/or occupational exposures that potentially could affect individual cancer risk was also unavailable. Nonetheless, this study is noteworthy from several perspectives, including the finding that it represents the most comprehensive investigation to date of cancer patterns among an American Indian tribal group ever completed. The investigators were able to match information from tribal registration records with cancer incidence records from the NYS Department of Health to identify incident cancers. Data on expected cases were generated by using the NYSCR, which is a population-based cancer registry.

Careful attention to matching procedures with the NYSCR, meticulous record keeping by the Tribal Clerk's office, and supplemental death certificate data all diminished the possibility of undercounting cancer cases among the SNI cohort. Nonetheless, it is possible that incident cancer cases may have been under reported to the NYSCR, especially before 1976, when the registry began to include cases from New York City. Nonetheless, there is no reason to believe that cancer cases among the Seneca are reported with a differential rate of accuracy compared with the general population.

Among the unique strengths of this study was the reliance on tribal roll books to identify enrolled members and the ability to conduct a comprehensive study over 50 years leading to the accrual of greater than 120,000 person-years of observation and overcoming the issue of limited sized populations, which challenge studies among American Indian tribal groups. It is important to acknowledge the finding that race (eg, tribal membership) may serve simply as a marker for other factors that have an impact on cancer risk, such as socioeconomic status, education, and employment, which may influence exposures to various cancer-inducing agents in both occupational and environmental settings, and health behaviors as well as access to and use of medical care.

Data from this study emphasize the need for individual tribal groups to take a more proactive role in addressing cancer prevention and control within their communities. Efforts should include smoking cessation and prevention, systematic tracking of adherence with a variety of cancer screening programs, access to comprehensive general cancer information, and health policy and environmental changes to support the adoption and maintenance of a healthy lifestyle.

Conflict of Interest Disclosures

The authors made no disclosures.

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